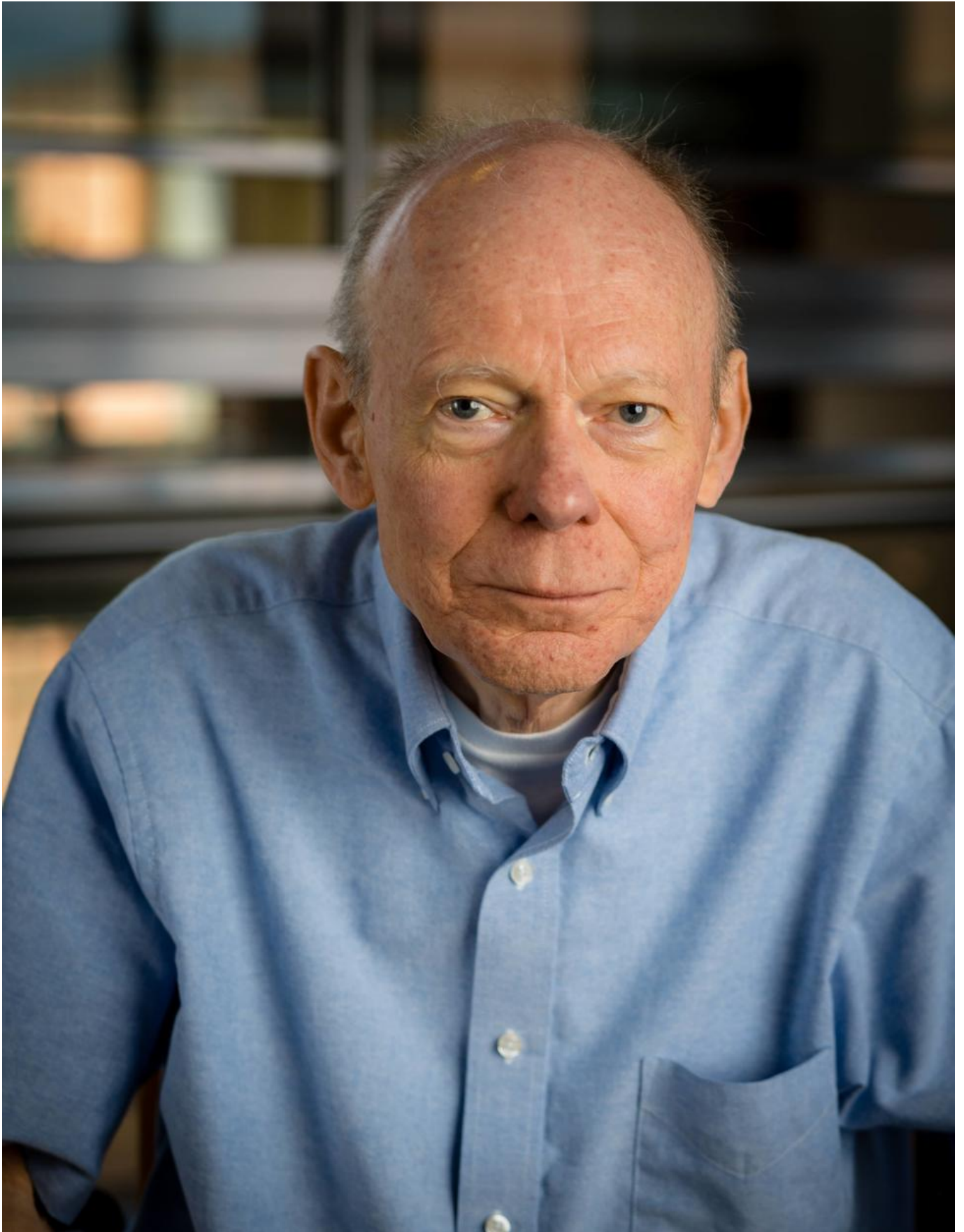


Reaching the Stars with a New Immunotherapy Drug



Credit: National Cancer Institute

In 2004, Allen's busy life as a NASA astrophysicist who modeled the evolution of stars was interrupted when he noticed enlarged lymph nodes on the side of his face and jaw. A biopsy revealed lymphoma. "It felt like being hit by a train. My first thought was that I might not see my teenage children grow up," Allen recalled.

The cancer initially was slow growing but became aggressive in 2012. Combination chemotherapy and an autologous stem cell transplant put it into remission.

Unfortunately, the cancer recurred in 2018. Having run out of standard treatment options, Allen jumped at the chance to join a phase 1 trial at the NIH Clinical Center, one of several sites testing a combination of rituximab (Rituxan) with Hu5F9-G4, a monoclonal antibody that blocks the CD47 protein on tumor cells. This treatment exposes tumor cells to destruction by the immune system. Basic and preclinical research by NCI-funded investigators at Stanford University helped lead to the drug's development.

Eight weeks after starting treatment, the now-retired Allen felt he had reached the stars again: His scans were negative. "NCI saved my life," he said. "I never expected to have such a quick response." Allen is a grandfather now and thrilled to be able to spend more time with his family.

Although Allen's trial was small, half the patients had a response to the treatment. Given its promising early results, Hu5F9-G4 is progressing in clinical trials for the treatment of multiple types of cancer.

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